

Capture the Flag

Last mission, you built reusable Angular components for the Student Portal.

Unfortunately, you hardcoded every piece of information directly into the HTML. Every student's profile looks identical, and announcements can only be changed by editing the HTML.

Your mission is to **bring the portal to life using Angular Interpolation (`{{ }}`)**.

Note from the instructor before you start:

These activities involve using your own logic. We are strictly to implement NOT TO USE AI to generate the logic for you. You may only use AI for RESEARCHING the syntax or UI, but NEVER THE LOGIC.

Consequence: Immediate 0 score in your grade for this activity.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/interpolation`



Capture

- Show your current Git branch.

Your branches					
Branch	Updated	Check status	Behind	Ahead	Pull request
<code>feature/interpolation</code>	3 minutes ago		0	2	...
<code>feature/components</code>	18 hours ago		0	2	...
Active branches					
Branch	Updated	Check status	Behind	Ahead	Pull request
<code>feature/interpolation</code>	3 minutes ago		0	2	...
<code>feature/components</code>	18 hours ago		0	2	...

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git checkout -b feature/interpolation
Switched to a new branch 'feature/interpolation'
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git branch
feature/components
* feature/interpolation
main
my-new-branch
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git push -u origin feature/interpolation
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/interpolation' on GitHub by visiting:
remote:   https://github.com/Jownieeey09/ITFE403A_Malate/pull/new/feature/interpolation
remote:
To https://github.com/Jownieeey09/ITFE403A_Malate.git
 * [new branch]      feature/interpolation -> feature/interpolation
branch 'feature/interpolation' set up to track 'origin/feature/interpolation'.
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git switch feature/interpolation
Already on 'feature/interpolation'
Your branch is up to date with 'origin/feature/interpolation'.
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project>
```

Flag 1 – Escape the Hardcoded Trap

Open your **Student Profile Card**.

Right now, your HTML probably looks something like this:

```
<h2>Juan Dela Cruz</h2>
<p>BSIT</p>
<p>3rd Year</p>
<p>2026-0001</p>
```

Your mission is to remove the hardcoded values.

Move them into your **TypeScript** component instead. (".ts")

Hint

```
{{ variableName }}
```

Your HTML should no longer contain actual student information. It should only reference variables.

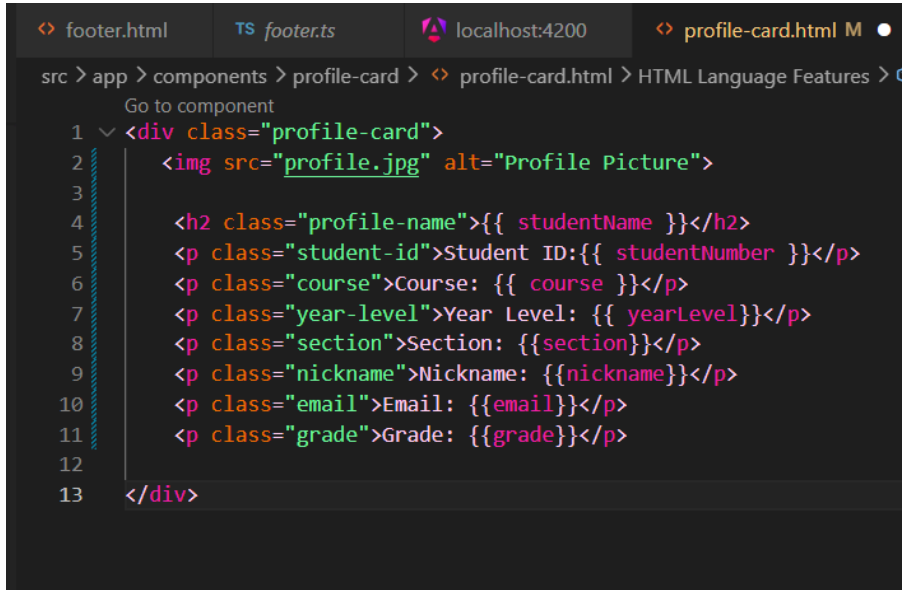
Your profile card should display:

- Student Name
- Student Number
- Course
- Year Level
- Section
- Nickname (all lowercase)

- Email
- Grade

Capture

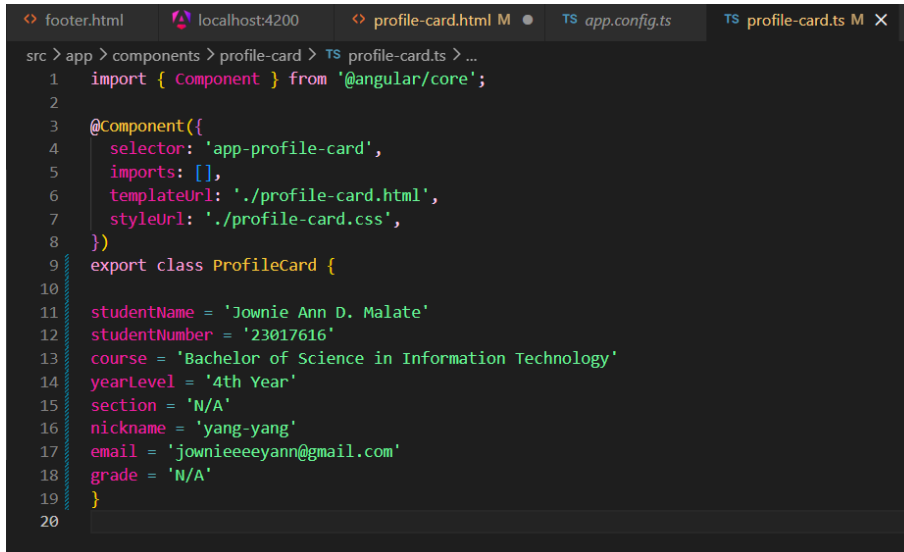
- HTML file
- TypeScript file
- Browser



```

src > app > components > profile-card > profile-card.html > HTML Language Features >
Go to component
1 <div class="profile-card">
2   
3
4   <h2 class="profile-name">{{ studentName }}</h2>
5   <p class="student-id">Student ID: {{ studentNumber }}</p>
6   <p class="course">Course: {{ course }}</p>
7   <p class="year-level">Year Level: {{ yearLevel }}</p>
8   <p class="section">Section: {{ section }}</p>
9   <p class="nickname">Nickname: {{ nickname }}</p>
10  <p class="email">Email: {{ email }}</p>
11  <p class="grade">Grade: {{ grade }}</p>
12
13 </div>

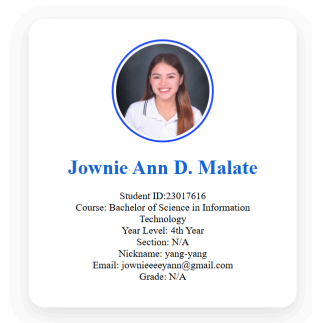
```



```

src > app > components > profile-card > TS profile-card.ts > ...
1 import { Component } from '@angular/core';
2
3 @Component({
4   selector: 'app-profile-card',
5   imports: [],
6   templateUrl: './profile-card.html',
7   styleUrls: ['./profile-card.css'],
8 })
9 export class ProfileCard {
10
11   studentName = 'Jownie Ann D. Malate'
12   studentNumber = '23017616'
13   course = 'Bachelor of Science in Information Technology'
14   yearLevel = '4th Year'
15   section = 'N/A'
16   nickname = 'yang-yang'
17   email = 'jownieeeeeyann@gmail.com'
18   grade = 'N/A'
19 }
20

```



Flag 2 – Dynamic Profile Card

Verify that **every** piece of information in your profile card now comes from TypeScript.

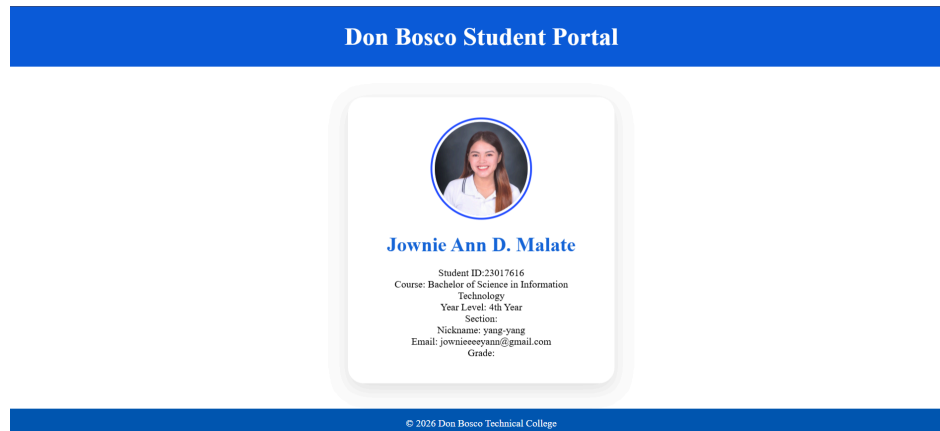
If changing a value in the TypeScript file updates the browser without touching the HTML, you've captured the flag.

Capture

- Updated TypeScript file with changed value
- Browser

```
<> footer.html  localhost:4200  <> profile-card.html 2, M  TS app.config.ts

src > app > components > profile-card > TS profile-card.ts > ...
1  import { Component } from '@angular/core';
2
3  @Component({
4    selector: 'app-profile-card',
5    imports: [],
6    templateUrl: './profile-card.html',
7    styleUrls: ['./profile-card.css'],
8  })
9  export class ProfileCard {
10
11    studentName = 'Jownie Ann D. Malate'
12    studentNumber = '23017616'
13    course = 'Bachelor of Science in Information Technology'
14    yearLevel = '4th Year'
15    nickname = 'yang-yang'
16    email = 'jownieeeeeyann@gmail.com'
17  }
18
```



Flag 3 – Dynamic Announcements

Your announcement component is also hardcoded.

Instead of writing directly in HTML, create variables for each announcement.

Example idea:

announcementTitle1
announcementBody1

announcementTitle2
announcementBody2

announcementTitle3
announcementBody3

Use interpolation so every announcement card gets its content from TypeScript.



Imagine these values came from a JSON file or API in the future.



Capture

- HTML file
- TypeScript file
- Browser

```
TS profile-card.ts M TS profile-card.spec.ts MyFirstAngularProject MyFirstAngularProject # profile-card.css announcement-card.html M X
src > app > components > announcement-card > announcement-card.html > HTML Language Features > div.announcement-card
Go to component
1 <div class="announcement-card">
2   <h2 class="announcement-title">{{ announcementTitle1 }}</h2>
3   <p class="Announcement-content">{{ announcementBody1 }}</p>
4 </div>
5
6 <div class="announcement-card">
7   <h2 class="announcement-title">{{ announcementTitle2 }}</h2>
8   <p class="Announcement-content">{{ announcementBody2 }}</p>
9 </div>
10
11 <div class="announcement-card">
12   <h2 class="announcement-title">{{ announcementTitle3 }}</h2>
13   <p class="Announcement-content">{{ announcementBody3 }}</p>
14 </div>
15
16 <div class="announcement-card">
17   <h2 class="announcement-title">{{ announcementTitle4 }}</h2>
18   <p class="Announcement-content">{{ announcementBody4 }}</p>
19 </div>
```

```
src > app > components > announcement-card > TS announcement-card.ts > Announcementcard
2
3 @Component({
4   selector: 'app-announcement-card',
5   standalone: true,
6   templateUrl: './announcement-card.html',
7   styleUrls: ['./announcement-card.css']
8 })
9 export class Announcementcard {
10   announcementTitle1 = '🔔 Enrollment Reminder';
11   announcementBody1 = 'Enrollment for the First Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.';
12
13   announcementTitle2 = '📷 Official ID Picture Taking';
14   announcementBody2 = 'Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students are required to wear their complete school uniform and present their registration form.';
15
16   announcementTitle3 = '🏆 Intramurals 2026';
17   announcementBody3 = 'Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.';
18
19 }
```

Latest Announcements

🔔 Enrollment Reminder

Enrollment for the First Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.

📷 Official ID Picture Taking

Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students are required to wear their complete school uniform and present their registration form.

🏆 Intramurals 2026

Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.

Flag 4 – Thinking Like Data

Now add 7 more cards to make the announcements a total of 10 cards.

Capture

- Show your ten announcement cards in Browser
- TypeScript file

```
src > app > components > announcement-card > TS announcement-card.ts > 📄 Announcementcard > 📄 announcementTitle1
1  import { Component } from '@angular/core';
2
3  @Component({
4    selector: 'app-announcement-card',
5    standalone: true,
6    templateUrl: './announcement-card.html',
7    styleUrls: ['./announcement-card.css']
8  })
9  export class Announcementcard {
10    announcementTitle1 = 'Enrollment Reminder';
11    announcementBody1 = 'Enrollment for the First Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.';
12
13    announcementTitle2 = 'Official ID Picture Taking';
14    announcementBody2 = 'Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM';
15
16    announcementTitle3 = 'Intramurals 2026';
17    announcementBody3 = 'Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September';
18
19    announcementTitle4 = 'SDP - CLUB MEMBER SCREENING';
20    announcementBody4 = 'Instructions for the Club Member Screening will be given after the holy mass.';
21
22    announcementTitle5 = 'CLASS SUSPENSION';
23    announcementBody5 = 'Due to inclement weather, all face-to-face classes are suspended today';
24
25    announcementTitle6 = 'BUWAN NG WIKA: BARRIO FIESTA CELEBRATION';
26    announcementBody6 = 'Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiestal!';
27
28    announcementTitle7 = 'EXAMINATION NOTICE';
29    announcementBody7 = 'Midterm examination will be on September 14, 2026';
30
31    announcementTitle8 = 'LIBRARY ADVISORY';
32    announcementBody8 = 'Students are reminded to return borrowed books on or before the due date to avoid overdue fines.';
33  }
```

```
    announcementTitle4 = 'SDP - CLUB MEMBER SCREENING';
    announcementBody4 = 'Instructions for the Club Member Screening will be given after the holy mass.';

    announcementTitle5 = 'CLASS SUSPENSION';
    announcementBody5 = 'Due to inclement weather, all face-to-face classes are suspended today';

    announcementTitle6 = 'BUWAN NG WIKA: BARRIO FIESTA CELEBRATION';
    announcementBody6 = 'Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiestal!';

    announcementTitle7 = 'EXAMINATION NOTICE';
    announcementBody7 = 'Midterm examination will be on September 14, 2026';

    announcementTitle8 = 'LIBRARY ADVISORY';
    announcementBody8 = 'Students are reminded to return borrowed books on or before the due date to avoid overdue fines.';

    announcementTitle9 = 'LOST AND FOUND';
    announcementBody9 = 'A black wallet was found near the school cafeteria. Owner may claim it at the SAO office by presenting proper identification.';

    announcementTitle10 = 'BOSCONIAN NOTICE';
    announcementBody10 = 'Get ready for an exciting and unforgettable evening as we celebrate Bosconian Night 2026!';
  }
```

Latest Announcements

Enrollment Reminder

Enrollment for the First Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.

Official ID Picture Taking

Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students are required to wear their complete school uniform and present their registration form.

Intramurals 2026

Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.

SDP - CLUB MEMBER SCREENING

Instructions for the Club Member Screening will be given after the holy mass.

BY: JACOB S. SANTIAGO

CLASS SUSPENSION

Due to inclement weather, all face-to-face classes are suspended today

BUWAN NG WIKA: BARRIO FIESTA CELEBRATION

Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiesta!

EXAMINATION NOTICE

Midterm examination will be on September 14, 2026

LIBRARY ADVISORY

Students are reminded to return borrowed books on or before the due date to avoid overdue fines.

LOST AND FOUND

A black wallet was found near the school cafeteria. Owner may claim it at the SAO office by presenting proper Identification and describing its contents.

BOSCONIAN NIGHT NOTICE

Get ready for an exciting and unforgettable evening as we celebrate Bosconian Night 2026!

Notice how your announcement variables are starting to repeat.

This is normal, but not ideal.

Question to think about (you don't need to answer, just think):

If there were 100 announcements, would creating 200 variables be a good idea?

(You'll solve this in a future lesson.)

Flag 5 – Academic Standing

Your profile already displays the student's grade.

Now make the portal smarter.

Below the grade, display:

- **Pass**
- **Fail**

depending on the grade.

Hint

Angular interpolation can evaluate expressions.

`{{ condition ? "True Value" : "False Value" }}`

Grading condition: A student can only pass if their grade is 1.0 to 3.0.

Capture

- Interpolation condition
- Browser

```
src > app > components > profile-card > <> profile-card.html > HTML Language Features > div.profile-card
1  <div class="profile-card">
4      <h2 class="profile-name">{{ studentName }}</h2>
5      <p class="student-id">Student ID: {{ studentNumber }}</p>
6      <p class="course">Course: {{ course }}</p>
7      <p class="year-level">Year Level: {{ yearLevel }}</p>
8      <p class="section">Section: {{ section }}</p>
9      <p class="nickname">Nickname: {{ nickname }}</p>
10     <p class="email">Email: {{ email }}</p>
11
12     <p class="grade">grade: {{ grade }}</p>
13     <b>{{ grade >= 1.0 && grade <= 3 ? 'Passed' : 'Failed' }}</b>
14
15 </div>
```



Jownie Ann D. Malate

Student ID:23017616
Course: Bachelor of Science in Information
Technology
Year Level: 4th Year
Section: N/A
Nickname: yang-yang
Email: jownieeeeeyann@gmail.com
grade: 1.5
Passed

Flag 6 – Formatting Text

Your student's nickname is stored in lowercase.

Example:

juan

Display it in uppercase without changing the value of the variable.

Hint

Interpolation can call simple methods.

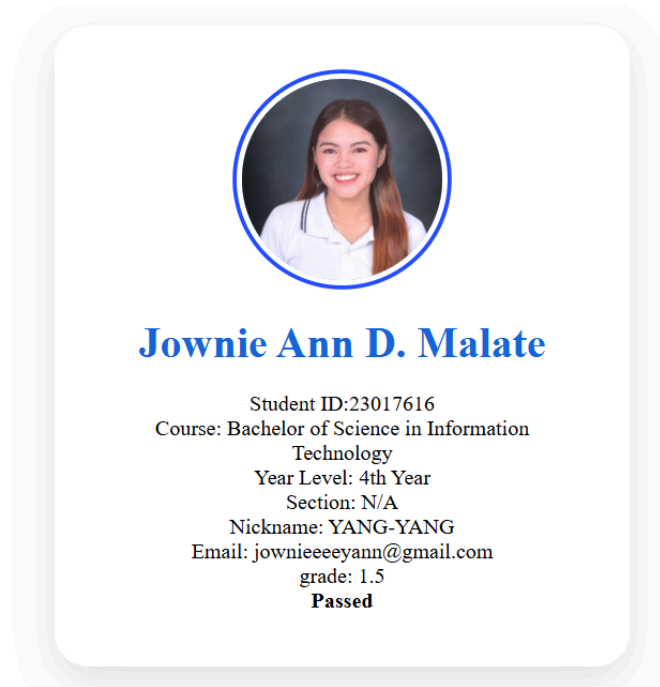
Expected:

JUAN

Capture

- Method used to display it in uppercase
- Browser

```
<p class="nickname">Nickname: {{nickname.toUpperCase()}}</p>
```



Flag 7 – School Email

Every student has the same format of email in Don Bosco.

Example:

juan.delacruz@dbtc-cebu.edu.ph

Instead of storing the full email, remove the “email” variable, then construct the email inside the template, by using the student’s name as variable, concatenated with the email suffix.

No extra variable should be created.

Capture

- Your logic to concatenate the name and email suffix using interpolation
- Browser

```
<p class="email">Email: {{ studentName.toLowerCase().replaceAll(searchValue: ' ', replaceValue: '') + '@dbtc-cebu.edu.ph' }}</p>
```



Jownie Ann D. Malate

Student ID:23017616
Course: Bachelor of Science in Information
Technology
Year Level: 4th Year
Section: N/A
Nickname: YANG-YANG
Email: jownieannd.malate@dbtc-cebu.edu.ph
grade: 1.5
Passed

Flag 8 – Copyright Footer

Every website needs a footer.

Create a variable:

`currentYear`

Display:

© 2026 Student Portal

using interpolation.

Example idea:

© {{ currentYear }}

 Capture

- Method used to display the currentYear without hardcoding “2026”
- Browser

```

1 <footer>
2   © {{ currentYear }} Don Bosco Technical College - Cebu
3
4 </footer>

```

```

})
export class Footer {
  currentYear = new Date(). getFullYear();
}

```

© 2026 Don Bosco Technical College - Cebu

Imagine the year changes from 2026 to 2027. If you hardcoded the year, you will have to change this value yearly. You need to have a function for that.

Flag 9 – Commit Your Progress

Final Flag Screenshot

- The entire completed browser UI for this activity
- Successful push to `feature/interpolation`

Don Bosco Student Portal



Jownie Ann D. Malate

Student ID: 23017616
 Course: Bachelor of Science in Information
 Technology
 Year Level: 4th Year
 Section: N/A
 Nickname: YANG-YANG
 Email: jownieand.malate@dbtc-cebu.edu.ph
 grade: 1.5
 Passed

Enrollment Reminder

Enrollment for the First Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.

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LIBRARY ADVISORY

Students are reminded to return borrowed books on or before the due date to avoid overdue fines.

LOST AND FOUND

A black wallet was found near the school cafeteria. Owner may claim it at the SAO office by presenting proper Identification and describing its contents.

BOSCONIAN NIGHT NOTICE

Get ready for an exciting and unforgettable evening as we celebrate Bosconian Night 2026!

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git status
    modified:   src/app/components/profile-card/profile-card.ts

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git add .
warning: in the working copy of 'src/app/components/announcement-card/announcement-card.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/announcement-card/announcement-card.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/footer/footer.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/footer/footer.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/profile-card/profile-card.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/profile-card/profile-card.ts', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git commit -m "feat: Activity: feature/interpolation"
[feature/interpolation 301dafd] feat: Activity: feature/interpolation
 8 files changed, 107 insertions(+), 57 deletions(-)
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git push origin feature/interpolation
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 8 threads
Compressing objects: 100% (16/16), done.
Writing objects: 100% (16/16), 2.95 KiB | 120.00 KiB/s, done.
Total 16 (delta 8), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (8/8), completed with 8 local objects.
To https://github.com/Jownieeee99/ITFE403A_Malate.git
   b411665..301dafd feature/interpolation -> feature/interpolation
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project>

```